RAI SCHOOL OF ENGINEERING

RAI UNIVERSITY

LAB MANUAL

Computer Network

Name: _____

Enrollment No.: _____

Programme / Semester: _____

Section / Batch: _____

Submitted To: _____

LIST OF EXPERIMENTS

Computer Network

Sr. No	Name of Experiment	Date	Page	Grade	Signature
1	Write down basic Networking commands.		3		
2	Configure Line (Bus) Topology in Cisco Packet Tracer.		6		
3	Configure Ring Topology in Cisco Packet Tracer.		8		
4	Configure Star Topology in Cisco Packet Tracer.		11		
5	Configure Mesh Topology in Cisco Packet Tracer.		13		
6	Configure Tree Topology in Cisco Packet Tracer.		16		
7	Configure Hybrid Topology in Cisco Packet Tracer.		18		
8	Write down steps to configure Local Area Network.		20		
9	Give Classification of Transmission Medias with diagram.		22		
10	Give IP address Classification.		24		

Experiment 1

Subject: Computer Network **Date:** _____

Problem Statement

Write down basic Networking commands.

Aim

To study and execute the basic networking commands used to check, configure and troubleshoot a network.

Theory

Networking commands are typed in the **Command Prompt** (Windows) or **Terminal** (Linux/macOS). They are used to view the network configuration of a computer, test whether another host can be reached, trace the path of packets, query the DNS and see the active connections.

Commands

Windows command	Linux equivalent	Purpose
`ipconfig`	`ifconfig / ip addr`	Shows the IP address, subnet mask and default gateway of every network adapter.
`ipconfig /all`	`ip addr, nmcli`	Shows full details: MAC address, DHCP server, DNS servers and lease time.
`ping <host>`	`ping <host>`	Tests reachability of a host by sending ICMP echo requests and shows the round-trip time and packet loss.
`tracert <host>`	`traceroute <host>`	Displays every router (hop) a packet passes through on the way to the destination.
`nslookup <name>`	`nslookup / dig`	Queries the DNS server to convert a domain name into an IP address and the reverse.
`netstat -an`	`netstat -an / ss -tuln`	Lists active connections, listening ports and their state.
`arp -a`	`arp -a / ip neigh`	Displays the ARP cache (IP address to MAC address mapping).
`hostname`	`hostname`	Displays the name of the computer.
`getmac`	`ip link`	Displays the MAC (physical) address of the network adapters.
`route print`	`route -n / ip route`	Displays the routing table.
`pathping <host>`	`mtr <host>`	Combines ping and tracert, and shows packet loss at each hop.
`ipconfig /release ipconfig /renew`	`dhclient -r dhclient`	Releases and renews the IP address given by a DHCP server.
`ipconfig /flushdns`	`resolvectl flush-caches`	Clears the DNS resolver cache.

Useful options

- `ping -t <host>` : ping continuously until stopped with Ctrl+C (Linux: plain ping runs continuously, use -c 4 for four packets).
- `ping -n 10 <host>` : send 10 requests (Linux: -c 10).
- `tracert -d <host>` : do not resolve names, so the trace is faster.
- `netstat -b` : show the program that owns each connection (needs administrator rights).
- `ping 127.0.0.1` : tests the TCP/IP stack of the local computer (loopback).

Sample Output

The addresses below are examples. The values on your computer will be different.

1. ipconfig

```
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.1.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
```

2. ping

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=24ms TTL=117
Reply from 8.8.8.8: bytes=32 time=23ms TTL=117
Reply from 8.8.8.8: bytes=32 time=25ms TTL=117
Reply from 8.8.8.8: bytes=32 time=24ms TTL=117
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 25ms, Average = 24ms
```

3. tracert

```
C:\>tracert example.com

Tracing route to example.com [203.0.113.10]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.1.1
  2     5 ms     4 ms     5 ms  10.10.0.1
  3    12 ms    11 ms    12 ms  172.16.8.5
  4    24 ms    25 ms    24 ms  203.0.113.10

Trace complete.
```

4. nslookup

```
C:\>nslookup example.com
Server:   router.local
Address:  192.168.1.1

Non-authoritative answer:
Name:     example.com
Address:  203.0.113.10
```

5. netstat -an

```
C:\>netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    192.168.1.10:50412     203.0.113.10:443       ESTABLISHED
  UDP    0.0.0.0:5353           *:*
```

6. arp -a and hostname

```
C:\>arp -a

Interface: 192.168.1.10 --- 0x4
  Internet Address      Physical Address      Type
  192.168.1.1           aa-bb-cc-dd-ee-ff     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static

C:\>hostname
LAB-PC-01
```

Result

The basic networking commands were studied and executed successfully.

Experiment 2

Subject: Computer Network **Date:** _____

Problem Statement

Configure Line (Bus) Topology in Cisco Packet Tracer.

Aim

To configure a Line (Bus) topology in Cisco Packet Tracer and verify communication between the computers.

Requirements

3 PCs (End Devices), 3 Hubs (Network Devices > Hubs > Hub-PT), Copper Straight-Through and Copper Cross-Over cables.

Theory

In a bus (line) topology all devices are connected to one common backbone cable. A signal sent by one device travels along the whole backbone and every device receives it, but only the device whose address matches accepts it. It needs little cable and is cheap, but a break in the backbone stops the whole network and the performance falls as more devices are added.

Packet Tracer has no coaxial bus cable. The shared backbone is therefore built with a **chain of hubs**, because a hub repeats every incoming signal to all its other ports, exactly as a shared bus does.

Topology Diagram

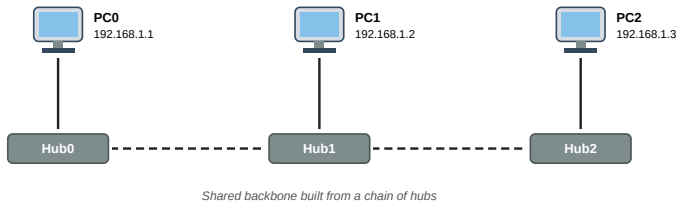


Fig. 2.1: Line (Bus) topology. Solid line: Copper Straight-Through. Dashed line: Copper Cross-Over.

Connections and IP Addressing

From	To	Cable Type
PC0	Hub0	Copper Straight-Through
PC1	Hub1	Copper Straight-Through
PC2	Hub2	Copper Straight-Through
Hub0	Hub1	Copper Cross-Over
Hub1	Hub2	Copper Cross-Over

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3

Subnet mask for all devices: 255.255.255.0. No default gateway is needed because all devices are in one network.

Procedure

1. Open Cisco Packet Tracer, start a new workspace, drag in the devices listed under Requirements and rename them as shown in the diagram.
2. Select the **Connections** (lightning) icon. For every row of the Connections table choose the stated cable, click the first device and pick a free port, then click the second device and pick a free port.
3. Click each PC, open **Desktop > IP Configuration**, select **Static** and enter the IP address and subnet mask from the IP Addressing table.
4. Wait until all the link lights on the cables turn green.
5. Open **PC0 > Desktop > Command Prompt** and run `ping 192.168.1.3`. Repeat from the other PCs.
6. (Optional) In **Simulation mode** use **Add Simple PDU** (closed envelope icon) between two PCs and press Play to watch the packet travel. Save the file as `bus_topology.pkt`.

Output

Ping from PC0 to PC2 passes through all three hubs and every request gets a reply:

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping may show one "Request timed out" while ARP finds the MAC address. The next attempt gives 0% loss.

Screenshot

Paste the Packet Tracer screenshot of the topology and the successful ping here

Advantages	Disadvantages
• Simple and easy to set up. • Needs less cable than other topologies, so it is cheap. • Easy to extend by adding another segment.	• A fault or break in the backbone stops the entire network. • Only one device can send at a time, so collisions increase as devices are added. • Faults are difficult to locate.

Result

The bus was configured in Cisco Packet Tracer and connectivity between the PCs was verified using the ping command.

Experiment 3

Subject: Computer Network **Date:** _____

Problem Statement

Configure Ring Topology in Cisco Packet Tracer.

Aim

To configure a Ring topology in Cisco Packet Tracer and verify communication between the computers.

Requirements

4 PCs (End Devices), 4 Switches (Network Devices > Switches > 2960), Copper Straight-Through and Copper Cross-Over cables.

Theory

In a ring topology every device is connected to exactly two neighbours, so the devices form a closed loop. Data travels around the ring from device to device until it reaches the destination. There is no central device and every device has an equal chance to send.

Here the ring is formed by 4 switches, each with one PC. A loop of switches could cause endless broadcast traffic, so the switches use the **Spanning Tree Protocol (STP)** to block one link of the ring automatically. If another link fails, STP opens the blocked link again, which shows the fault tolerance of the ring.

Topology Diagram

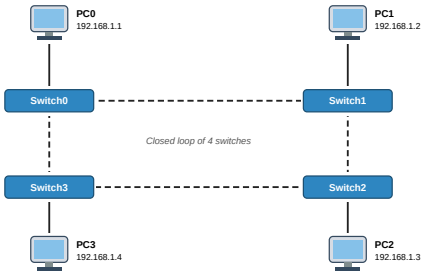


Fig. 3.1: Ring topology of four switches. Solid line: Copper Straight-Through. Dashed line: Copper Cross-Over.

Connections and IP Addressing

From	To	Cable Type
PC0	Switch0	Copper Straight-Through
PC1	Switch1	Copper Straight-Through
PC2	Switch2	Copper Straight-Through
PC3	Switch3	Copper Straight-Through
Switch0	Switch1	Copper Cross-Over
Switch1	Switch2	Copper Cross-Over
Switch2	Switch3	Copper Cross-Over
Switch3	Switch0	Copper Cross-Over

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4

Subnet mask for all devices: 255.255.255.0. No default gateway is needed because all devices are in one network.

Procedure

1. Open Cisco Packet Tracer, start a new workspace, drag in the devices listed under Requirements and rename them as shown in the diagram.
2. Select the **Connections** (lightning) icon. For every row of the Connections table choose the stated cable, click the first device and pick a free port, then click the second device and pick a free port.
3. Click each PC, open **Desktop > IP Configuration**, select **Static** and enter the IP address and subnet mask from the IP Addressing table.
4. Wait about 30 seconds. The switch ports first show amber and then turn green. One port of the ring stays **amber**: this is the port blocked by STP to break the loop.
5. (Optional) Click Switch0, open the **CLI** tab and run `enable` and then `show spanning-tree` to see the root bridge and which port is in the BLK (blocking) state.
6. (Optional) Delete one cable of the ring, wait for STP to converge, and ping again. The ping still succeeds because the blocked link takes over.
7. Open **PC0 > Desktop > Command Prompt** and run `ping 192.168.1.3`. Repeat from the other PCs.
8. (Optional) In **Simulation mode** use **Add Simple PDU** (closed envelope icon) between two PCs and press Play to watch the packet travel. Save the file as `ring_topology.pkt`.

Output

Ping from PC0 to PC2 is successful, and all other PCs are also reachable:

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping may show one "Request timed out" while ARP finds the MAC address. The next attempt gives 0% loss.

Note: because STP blocks one link, the 4 switches are logically connected as a line, but the physical topology is a ring.

Screenshot

Paste the Packet Tracer screenshot of the topology and the successful ping here

Advantages	Disadvantages
• Data flow is orderly and no central device is needed. • Every device has equal access to the network. • With STP or a dual ring, the network survives a single link failure.	• A break in a plain ring can stop the whole network. • Adding or removing a device disturbs the ring. • Faults are harder to find than in a star.

Result

The ring was configured in Cisco Packet Tracer and connectivity between the PCs was verified using the ping command.

Experiment 4

Subject: Computer Network **Date:** _____

Problem Statement

Configure Star Topology in Cisco Packet Tracer.

Aim

To configure a Star topology in Cisco Packet Tracer and verify communication between the computers.

Requirements

5 PCs (End Devices), 1 Switch (Network Devices > Switches > 2960), Copper Straight-Through cables.

Theory

In a star topology every device is connected to a central device, a switch or a hub, by its own cable. All communication passes through the central device. It is easy to install and manage, a cable fault affects only one device, and new devices can be added without disturbing the network. The disadvantage is that the whole network stops if the central switch fails.

Topology Diagram

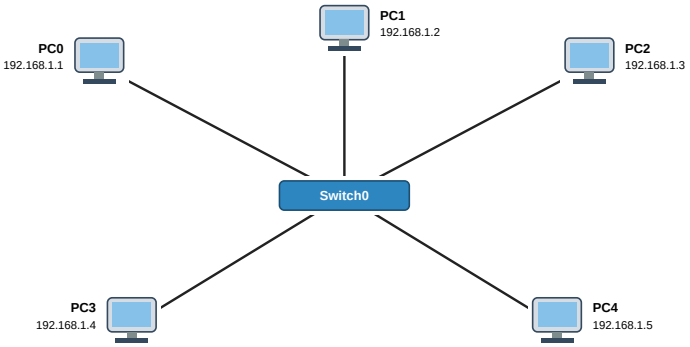


Fig. 4.1: Star topology. All connections are Copper Straight-Through.

Connections and IP Addressing

From	To	Cable Type
PC0	Switch0	Copper Straight-Through
PC1	Switch0	Copper Straight-Through
PC2	Switch0	Copper Straight-Through
PC3	Switch0	Copper Straight-Through
PC4	Switch0	Copper Straight-Through

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4
PC4	192.168.1.5

Subnet mask for all devices: 255.255.255.0. No default gateway is needed because all devices are in one network.

Procedure

1. Open Cisco Packet Tracer, start a new workspace, drag in the devices listed under Requirements and rename them as shown in the diagram.
2. Select the **Connections** (lightning) icon. For every row of the Connections table choose the stated cable, click the first device and pick a free port, then click the second device and pick a free port.
3. Click each PC, open **Desktop > IP Configuration**, select **Static** and enter the IP address and subnet mask from the IP Addressing table.

4. Wait until all the link lights on the cables turn green (the switch ports are amber for a few seconds at first).
5. Open **PC0 > Desktop > Command Prompt** and run `ping 192.168.1.5`. Repeat from the other PCs.
6. (Optional) In **Simulation mode** use **Add Simple PDU** (closed envelope icon) between two PCs and press Play to watch the packet travel. Save the file as `star_topology.pkt`.

Output

Ping from PC0 to PC4 through the central switch is successful:

```
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping may show one "Request timed out" while ARP finds the MAC address. The next attempt gives 0% loss.

Screenshot

Paste the Packet Tracer screenshot of the topology and the successful ping here

Advantages	Disadvantages
• Easy to install, expand and manage. • A failure of one cable or device does not affect the others. • Faults are easy to detect and isolate.	• If the central switch fails, the whole network fails. • Needs more cable than a bus. • The cost of the central device is added.

Result

The star was configured in Cisco Packet Tracer and connectivity between the PCs was verified using the ping command.

Experiment 5

Subject: Computer Network **Date:** _____

Problem Statement

Configure Mesh Topology in Cisco Packet Tracer.

Aim

To configure a (full) Mesh topology in Cisco Packet Tracer and verify communication between the computers.

Requirements

4 PCs (End Devices), 4 Switches (Network Devices > Switches > 2960), Copper Straight-Through and Copper Cross-Over cables.

Theory

In a full mesh topology every device has a direct link to every other device. For *n* devices the number of links is *n(n-1)/2*, so 4 switches need 6 links. Many paths exist, so the network is very reliable and traffic can be re-routed if a link fails. The cabling, however, is costly and complex.

Because many loops exist, the switches run **STP**. A spanning tree of 4 switches uses only 3 links, so STP keeps 3 of the 6 links forwarding and blocks the others. The blocked links act as backups and become active if a forwarding link fails.

Topology Diagram

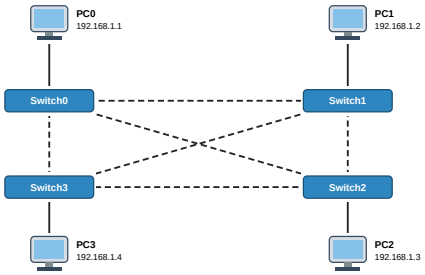


Fig. 5.1: Full mesh of four switches (6 links). Solid line: Copper Straight-Through. Dashed line: Copper Cross-Over.

Connections and IP Addressing

From	To	Cable Type
PC0	Switch0	Copper Straight-Through
PC1	Switch1	Copper Straight-Through
PC2	Switch2	Copper Straight-Through
PC3	Switch3	Copper Straight-Through
Switch0	Switch1	Copper Cross-Over
Switch0	Switch2	Copper Cross-Over
Switch0	Switch3	Copper Cross-Over
Switch1	Switch2	Copper Cross-Over
Switch1	Switch3	Copper Cross-Over
Switch2	Switch3	Copper Cross-Over

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4

Subnet mask for all devices: 255.255.255.0. No default gateway is needed because all devices are in one network.

Procedure

1. Open Cisco Packet Tracer, start a new workspace, drag in the devices listed under Requirements and rename them as shown in the diagram.
2. Select the **Connections** (lightning) icon. For every row of the Connections table choose the stated cable, click the first device and pick a free port, then click the second device and pick a free port.
3. Click each PC, open **Desktop > IP Configuration**, select **Static** and enter the IP address and subnet mask from the IP Addressing table.
4. Wait about 30 seconds until the ports settle. Some inter-switch ports stay **amber**: these links are blocked by STP.
5. (Optional) Open the CLI of any switch and run `enable` and `show spanning-tree` to see the forwarding (FWD) and blocking (BLK) ports.
6. Open **PC0 > Desktop > Command Prompt** and run `ping 192.168.1.4`. Repeat from the other PCs.
7. (Optional) In **Simulation mode** use **Add Simple PDU** (closed envelope icon) between two PCs and press Play to watch the packet travel. Save the file as `mesh_topology.pkt`.

Output

Ping from PC0 to PC3 is successful:

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping may show one "Request timed out" while ARP finds the MAC address. The next attempt gives 0% loss.

Screenshot

Paste the Packet Tracer screenshot of the topology and the successful ping here

Advantages	Disadvantages
• Very reliable: many alternate paths exist. • Each link carries only the traffic of the two devices it joins. • A fault in one link does not stop the network.	• Very expensive because of the large amount of cable and ports. • Installation and configuration are complex. • The number of links grows as n(n-1)/2.

Result

The mesh was configured in Cisco Packet Tracer and connectivity between the PCs was verified using the ping command.

Experiment 6

Subject: Computer Network **Date:** _____

Problem Statement

Configure Tree Topology in Cisco Packet Tracer.

Aim

To configure a Tree topology in Cisco Packet Tracer and verify communication between the computers.

Requirements

4 PCs (End Devices), 3 Switches (Network Devices > Switches > 2960), Copper Straight-Through and Copper Cross-Over cables.

Theory

A tree topology is a hierarchy. A root (core) switch is connected to second-level switches, and each of these is connected to end devices. It combines the features of star and bus topologies. The network is easy to expand by adding new branches, and a fault is limited to its branch. If the root switch fails, the branches are separated from each other.

Topology Diagram

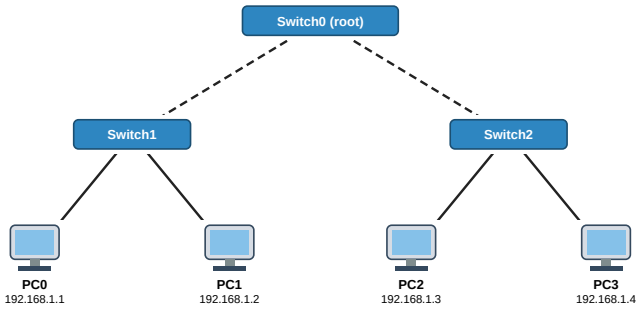


Fig. 6.1: Tree topology. Solid line: Copper Straight-Through. Dashed line: Copper Cross-Over.

Connections and IP Addressing

From	To	Cable Type
PC0	Switch1	Copper Straight-Through
PC1	Switch1	Copper Straight-Through
PC2	Switch2	Copper Straight-Through
PC3	Switch2	Copper Straight-Through
Switch0 (root)	Switch1	Copper Cross-Over
Switch0 (root)	Switch2	Copper Cross-Over

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4

Subnet mask for all devices: 255.255.255.0. No default gateway is needed because all devices are in one network.

Procedure

1. Open Cisco Packet Tracer, start a new workspace, drag in the devices listed under Requirements and rename them as shown in the diagram.
2. Select the **Connections** (lightning) icon. For every row of the Connections table choose the stated cable, click the first device and pick a free port, then click the second device and pick a free port.
3. Click each PC, open **Desktop > IP Configuration**, select **Static** and enter the IP address and subnet mask from the IP Addressing table.
4. Wait until all the link lights turn green.
5. Open **PC0 > Desktop > Command Prompt** and run `ping 192.168.1.4`. Repeat from the other PCs.
6. (Optional) In **Simulation mode** use **Add Simple PDU** (closed envelope icon) between two PCs and press Play to watch the packet travel. Save the file as `tree_topology.pkt`.

Output

Ping from PC0 (under Switch1) to PC3 (under Switch2) goes through the root switch and is successful:

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping may show one "Request timed out" while ARP finds the MAC address. The next attempt gives 0% loss.

Screenshot

Paste the Packet Tracer screenshot of the topology and the successful ping here

Advantages	Disadvantages
• Easy to expand by adding branches. • Hierarchical design is easy to manage and troubleshoot. • A fault in one branch does not affect the other branches.	• If the root switch fails, the branches cannot communicate. • Needs more cabling than a bus. • Configuration grows complex as the tree grows.

Result

The tree was configured in Cisco Packet Tracer and connectivity between the PCs was verified using the ping command.

Experiment 7

Subject: Computer Network **Date:** _____

Problem Statement

Configure Hybrid Topology in Cisco Packet Tracer.

Aim

To configure a Hybrid topology (Star-Ring) in Cisco Packet Tracer and verify communication between the computers.

Requirements

6 PCs (End Devices), 3 Switches (Network Devices > Switches > 2960), Copper Straight-Through and Copper Cross-Over cables.

Theory

A hybrid topology combines two or more different basic topologies. In this experiment a **ring** (the three switches joined in a loop) is combined with **stars** (each switch is the centre of a star with two PCs). It gives the redundancy of a ring and the easy management of a star, and it can be extended easily.

STP blocks one of the three ring links to prevent a loop, and re-opens it if another ring link fails.

Topology Diagram

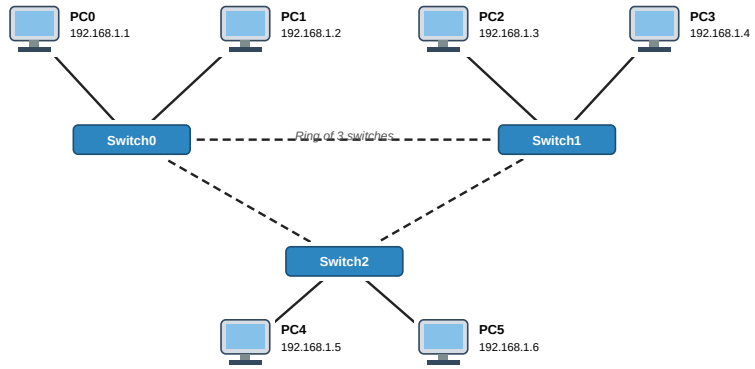


Fig. 7.1: Hybrid (Star-Ring) topology. Solid line: Copper Straight-Through. Dashed line: Copper Cross-Over.

Connections and IP Addressing

From	To	Cable Type
PC0, PC1	Switch0	Copper Straight-Through
PC2, PC3	Switch1	Copper Straight-Through
PC4, PC5	Switch2	Copper Straight-Through
Switch0	Switch1	Copper Cross-Over
Switch1	Switch2	Copper Cross-Over
Switch2	Switch0	Copper Cross-Over

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4
PC4	192.168.1.5
PC5	192.168.1.6

Subnet mask for all devices: 255.255.255.0. No default gateway is needed because all devices are in one network.

Procedure

1. Open Cisco Packet Tracer, start a new workspace, drag in the devices listed under Requirements and rename them as shown in the diagram.
2. Select the **Connections** (lightning) icon. For every row of the Connections table choose the stated cable, click the first device and pick a free port, then click the second device and pick a free port.
3. Click each PC, open **Desktop > IP Configuration**, select **Static** and enter the IP address and subnet mask from the IP Addressing table.
4. Wait about 30 seconds. One ring port stays **amber** because STP blocks one of the three ring links.
5. Open **PC0 > Desktop > Command Prompt** and run `ping 192.168.1.6`. Repeat from the other PCs.
6. (Optional) In **Simulation mode** use **Add Simple PDU** (closed envelope icon) between two PCs and press Play to watch the packet travel. Save the file as `hybrid_topology.pkt`.

Output

Ping from PC0 (on Switch0) to PC5 (on Switch2) is successful:

```
C:\>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping may show one "Request timed out" while ARP finds the MAC address. The next attempt gives 0% loss.

Screenshot

Paste the Packet Tracer screenshot of the topology and the successful ping here

Advantages	Disadvantages
• Flexible: each part can use the topology that suits it best. • Reliable and easy to expand. • Faults are isolated to a part of the network.	• Design and installation are complex. • Costlier than a single topology. • Needs more careful management.

Result

The hybrid was configured in Cisco Packet Tracer and connectivity between the PCs was verified using the ping command.

Experiment 8

Subject: Computer Network **Date:** _____

Problem Statement

Write down steps to configure Local Area Network.

Aim

To write down the steps to configure a Local Area Network (LAN).

Requirements

Two or more computers with a network card (NIC), a switch (or hub), UTP straight-through cables (Cat5e or Cat6) with RJ-45 connectors.

Theory

A LAN connects computers within a small area such as a laboratory, office or building, usually through a switch. For computers to communicate directly they must be in the **same network**, which means the same network part of the IP address and the same subnet mask.

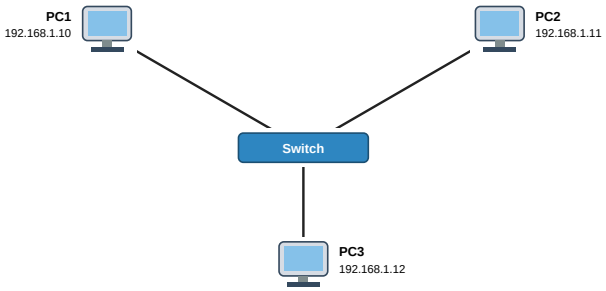


Fig. 8.1: A simple LAN with a switch and three computers

Computer	IP Address	Subnet Mask
PC1	192.168.1.10	255.255.255.0
PC2	192.168.1.11	255.255.255.0
PC3	192.168.1.12	255.255.255.0

Steps

1. **Connect the hardware.** Connect each computer to a port of the switch with a UTP straight-through cable. Check that the link lights of the network card and the switch port are on.
2. **Open the adapter settings.** Windows 10: Control Panel > Network and Internet > Network and Sharing Center > Change adapter settings. Windows 11: Settings > Network & internet > Ethernet. Right-click the **Ethernet** adapter and choose **Properties**.
3. **Set the IP address.** Select **Internet Protocol Version 4 (TCP/IPv4)** and click Properties. Choose **Use the following IP address** and enter a different address on each computer, for example `192.168.1.10` with the subnet mask `255.255.255.0`. The default gateway is needed only if the LAN is connected to the Internet through a router.
4. **Give each computer a name and workgroup.** Settings > System > About > Rename this PC (advanced). Use a different computer name on each PC and the same workgroup (for example WORKGROUP) on all of them.

5. **Turn on sharing.** Control Panel > Network and Sharing Center > Advanced sharing settings. Turn on **Network discovery** and **File and printer sharing**.
6. **Test the connection.** Open Command Prompt and run `ping 192.168.1.11`. Replies show that the LAN works. Windows Firewall blocks ping by default, so if the request times out, allow the rule *File and Printer Sharing (Echo Request - ICMPv4-In)* or turn the firewall off for the test.
7. **Share a folder.** Right-click the folder > Properties > Sharing > Share, and select the users. Other computers can open it by typing `\\192.168.1.10` in the File Explorer address bar.
8. **Share a printer (optional).** Printer Properties > Sharing > Share this printer.

Same LAN in Cisco Packet Tracer

1. Place one switch and three PCs and join each PC to the switch with a Copper Straight-Through cable.
2. On every PC open Desktop > IP Configuration, choose Static and enter the address and mask from the table.
3. Open Command Prompt on PC1 and ping PC2 and PC3.

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Troubleshooting

Problem	Likely cause and fix
No link light on the port	Faulty cable or loose RJ-45 connector. Re-seat or replace the cable.
Ping: Destination host unreachable	The computers are in different networks. Check the IP address and subnet mask.
Ping: Request timed out	Firewall blocks ICMP, or the other computer is off. Allow ICMP echo or turn off the firewall for the test.
IP address conflict warning	Two computers have the same IP address. Give each one a unique address.

Result

A Local Area Network was configured and connectivity between the computers was verified.

Experiment 9

Subject: Computer Network **Date:** _____

Problem Statement

Give Classification of Transmission Medias with diagram.

Aim

To study the classification of transmission media with a diagram.

Theory

Transmission media is the physical path through which data travels from the sender to the receiver. It is divided into two main groups. In **guided (wired) media** the signal travels along a physical cable. In **unguided (wireless) media** the signal travels through air or space as electromagnetic waves.

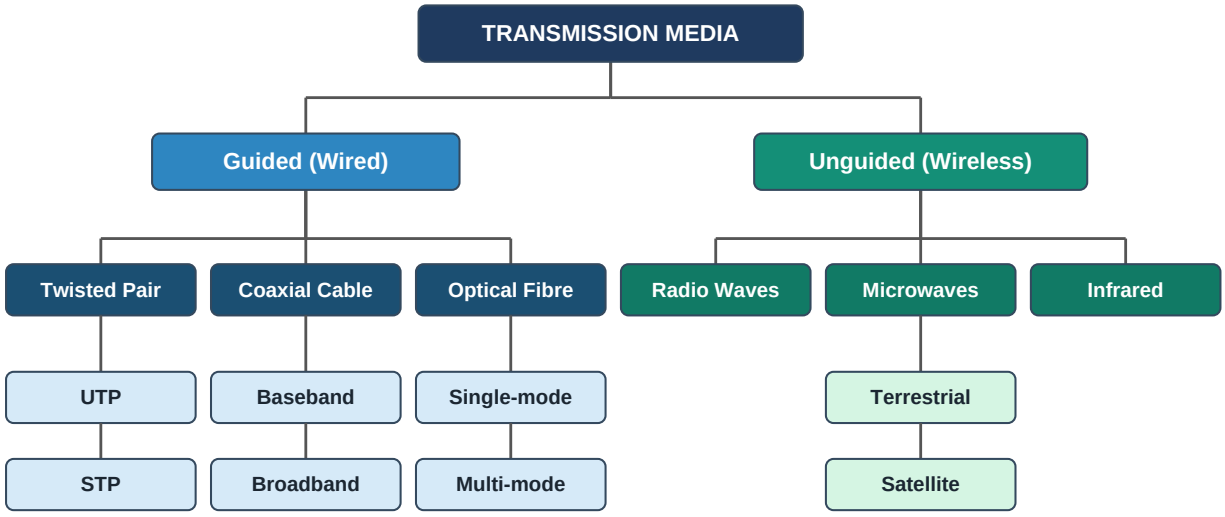


Fig. 9.1: Classification of transmission media

A. Guided Media

Medium	Description	Features
Twisted pair	Pairs of insulated copper wires twisted together to reduce interference. **UTP** (Unshielded Twisted Pair) is used in LAN cabling. **STP** (Shielded Twisted Pair) has a metal shield for extra protection from noise.	Cheap, flexible and easy to install. Range about 100 m per segment. UTP is affected by noise.
Coaxial cable	A central copper conductor surrounded by insulation, a metal shield and an outer jacket. **Baseband** carries one digital signal (old Ethernet 10Base2 and 10Base5). **Broadband** carries many signals at different frequencies (cable TV).	Better shielding and bandwidth than twisted pair, but thicker and costlier.
Optical fibre	Carries data as pulses of light through a glass or plastic core. **Single-mode** has a very thin core and one light path, for very long distances. **Multi-mode** has a thicker core and many light paths, for shorter distances such as buildings and campuses.	Very high speed and bandwidth, no electromagnetic interference, secure and long range, but expensive and harder to install.

Comparison of Guided Media

Medium	Speed / Bandwidth	Typical distance	Cost	Noise immunity
UTP	Up to 10 Gbps (Cat6a)	About 100 m	Low	Low
STP	Up to 10 Gbps	About 100 m	Medium	Medium
Coaxial	Medium	185 m (10Base2), 500 m (10Base5)	Medium	Good
Optical fibre	Very high (Gbps to Tbps)	Kilometres (tens of km for single-mode)	High	Excellent (immune to EMI)

Common UTP categories

Category	Maximum speed	Used for
Cat5e	1 Gbps	Gigabit Ethernet, 100 m
Cat6	1 Gbps (10 Gbps up to 55 m)	Gigabit and short 10 Gigabit links
Cat6a	10 Gbps	10 Gigabit Ethernet, 100 m

B. Unguided Media

Medium	Frequency range	Description
Radio waves	3 kHz to 1 GHz	Omnidirectional: they spread in all directions and pass through walls. Used in radio, TV, mobile phones, Wi-Fi and Bluetooth.
Microwaves	1 GHz to 300 GHz	Unidirectional and need line of sight. **Terrestrial** microwave uses towers; **satellite** microwave uses a satellite as a relay for very long distances.
Infrared	300 GHz to 400 THz	Short range, cannot pass through walls. Used in TV remotes and short-range device-to-device links.

Result

The classification of transmission media was studied with a diagram.

Experiment 10

Subject: Computer Network **Date:** _____

Problem Statement

Give IP address Classification.

Aim

To study the classification of IP addresses.

Theory

An IPv4 address is a 32-bit number written as four decimal numbers (octets) separated by dots, for example `192.168.1.10`. Each octet has 8 bits and a value from 0 to 255. The address has a **network part** (identifies the network) and a **host part** (identifies the device in that network). In classful addressing the IPv4 address space is divided into five classes, A to E, decided by the first bits of the first octet.

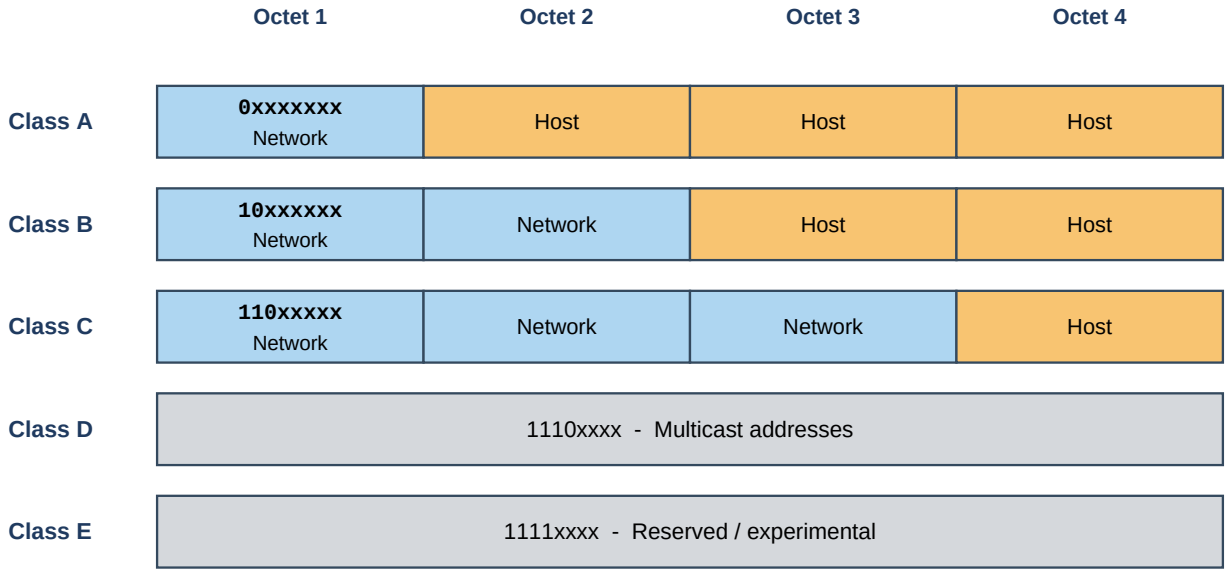


Fig. 10.1: Structure of the IPv4 address classes

Classification Table

Class	First bits	First octet	Default mask	Format	Networks	Hosts per network
A	0	1 - 126	255.0.0.0 (/8)	N.H.H.H	126	16,777,214
B	10	128 - 191	255.255.0.0 (/16)	N.N.H.H	16,384	65,534
C	110	192 - 223	255.255.255.0 (/24)	N.N.N.H	2,097,152	254
D	1110	224 - 239	Not defined	Multicast	-	-
E	1111	240 - 255	Not defined	Reserved / experimental	-	-

N = network bits, H = host bits. Hosts per network = $2^{(\text{host bits})}$ - 2, because the first address (all host bits 0) is the **network address** and the last address (all host bits 1) is the **broadcast address**. Class A: 2^{24} - 2 = 16,777,214. Class B: 2^{16} - 2 = 65,534. Class C: 2^{8} - 2 = 254.

Class A has 128 network numbers (2^{7}), but 0.x.x.x and 127.x.x.x are reserved, so 126 are usable. Class B has 2^{14} = 16,384 networks and Class C has 2^{21} = 2,097,152 networks.

Private IP Address Ranges

These addresses are used inside local networks and are not routed on the public Internet.

Class	Range	CIDR block
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

Special Addresses

Address	Meaning
127.0.0.0/8 (e.g. 127.0.0.1)	Loopback: used to test the TCP/IP software of the local computer.
0.0.0.0	Unspecified (this host) or the default route.
255.255.255.255	Limited broadcast to all hosts on the local network.
169.254.0.0/16	APIPA: automatic private address used when no DHCP server answers.

Examples

IP address	Class	Network address	Broadcast address	Usable hosts
10.2.3.4	A	10.0.0.0	10.255.255.255	10.0.0.1 - 10.255.255.254
172.16.5.4	B	172.16.0.0	172.16.255.255	172.16.0.1 - 172.16.255.254
192.168.1.10	C	192.168.1.0	192.168.1.255	192.168.1.1 - 192.168.1.254
224.0.0.5	D	Multicast address	-	-

Note: classful addressing has been largely replaced by **CIDR** (Classless Inter-Domain Routing), which allows any prefix length such as /22. The classes are still studied as the basic concept of IP addressing.

Result

The classification of IP addresses was studied.